

Homotopy Pushouts and Singular Chains

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1 Homotopy Pushouts of Spaces

The homotopy in a square is part of its gluing data. The double mapping cylinder records that homotopy as an interval over each point of the common space.

1.1 The Double Mapping Cylinder

Write $I = [0, 1]$ and let Top denote the category of topological spaces and continuous maps. All spaces in this lecture are objects of Top . For a space W , write $\text{pr}_W : W \rightarrow *$ for its unique map to a point.

Definition 1.1. For continuous maps $f : W \rightarrow X$ and $g : W \rightarrow Y$, the *double mapping cylinder* is

$$\text{Cyl}(f, g) = (X + (W \times I) + Y) / ((w, 0) \sim f(w), (w, 1) \sim g(w)).$$

Its canonical maps are $\iota_X : X \rightarrow \text{Cyl}(f, g)$ and $\iota_Y : Y \rightarrow \text{Cyl}(f, g)$. The map $\eta : W \times I \rightarrow \text{Cyl}(f, g)$, given by $\eta(w, t) = [w, t]$, is a homotopy from $\iota_X f$ to $\iota_Y g$.

Proposition 1.2. For a span $X \xleftarrow{f} W \xrightarrow{g} Y$ of topological spaces and a topological space Z , maps $\text{Cyl}(f, g) \rightarrow Z$ correspond bijectively to maps $i : X \rightarrow Z$, $j : Y \rightarrow Z$, and homotopies $h : W \times I \rightarrow Z$ satisfying

$$h(w, 0) = if(w), \quad h(w, 1) = jg(w).$$

The correspondence is natural in Z .

Proof. A map on the disjoint union in Definition 1.1 is specified by its restrictions i, h, j . It factors through the quotient precisely when the restrictions agree at the identified endpoints. These are the displayed conditions. The quotient topology makes the induced map continuous, and surjectivity of the quotient map gives uniqueness. Postcomposition preserves all three restrictions, proving naturality. $\square$

Definition 1.3. A *homotopy pushout square* of spaces consists of continuous maps and a specified homotopy $h : if \simeq jg$,

$$\begin{array}{ccc} W & \xrightarrow{f} & X \\ g \downarrow & & \downarrow i \\ Y & \xrightarrow{j} & Z, \end{array} \quad h : W \times I \rightarrow Z,$$

such that the induced comparison map

$$\theta_h : \text{Cyl}(f, g) \rightarrow Z, \quad \theta_h|_X = i, \quad \theta_h|_Y = j, \quad \theta_h([w, t]) = h(w, t),$$

is a homotopy equivalence. Proposition 1.2 defines this map.

For a strictly commutative square the constant homotopy is available, but the homotopy used in Definition 1.3 is still specified. Changing it changes the comparison map.

1.2 Suspensions, Mapping Cones, and Gluing

The double mapping cylinder includes suspension and cone constructions. Cofibrations explain when ordinary pushouts already have this homotopy type.

Example 1.4. For the maps pr_W at both ends, $\text{Cyl}(\text{pr}_W, \text{pr}_W)$ is the unreduced suspension SW . It is obtained from $W \times I$ by the identifications $(w, 0) \sim *_0$ and $(w, 1) \sim *_1$, where $*_0$ and $*_1$ are distinct vertices. When $W = S^0$ consists of two points, the suspension is a circle. The ordinary pushout of $* \xleftarrow{\text{pr}_{S^0}} S^0 \xrightarrow{\text{pr}_{S^0}} *$ is a point, so ordinary pushouts and homotopy pushouts can have different homotopy types.

Example 1.5. For a continuous map $f : W \rightarrow X$, its *mapping cone* is

$$\text{Cone}(f) = \text{Cyl}(f, \text{pr}_W).$$

The square with vertices $W, X, *, \text{Cone}(f)$ is a homotopy pushout with its cylinder homotopy. If W is nonempty, this is the usual quotient of $X + (W \times I)$ with the identifications $(w, 0) \sim f(w)$ and $(w, 1) \sim *$, where $*$ is the cone vertex. If $W = \emptyset$, the mapping cone is $X + *$.

Definition 1.6. A continuous map $f : X \rightarrow Y$ is a *Hurewicz cofibration* if it has the homotopy extension property for every target space: a map $h : Y \rightarrow Z$ and a homotopy $\alpha : X \times I \rightarrow Z$ starting at hf extend to a homotopy $\beta : Y \times I \rightarrow Z$ starting at h , with $\beta(f(x), t) = \alpha(x, t)$. For a subspace inclusion this is the extension problem

$$\begin{array}{ccc} (Y \times \{0\}) & \xrightarrow{X \times \{0\}} & (X \times I) \longrightarrow Z \\ \downarrow & \nearrow \beta & \\ Y \times I & & \end{array}$$

for every map from the upper-left space.

For a Hurewicz cofibration $W \hookrightarrow X$, the canonical map defined by $[w, t] \mapsto [f(w)] = [g(w)]$ gives a homotopy equivalence $\text{Cyl}(f, g) \rightarrow X \overset{+}{\underset{W}{Y}}$. The homotopy extension proof is the double-mapping-cylinder lemma: May, Chapter 10, §7, p. 80. In particular, cell attachments and gluing CW complexes along a common subcomplex give homotopy pushout squares.

2 Homotopy Pushouts of Complexes

For complexes, a degree-one map records the homotopy in a square. A mapping cone assembles the two complexes and the shifted common complex, and its homology gives the gluing exact sequence.

2.1 The Comparison Map and Its Signs

Let $\text{Ch}(\mathbb{Z})$ denote the category of homologically graded complexes of abelian groups and chain maps. A complex $A_\bullet$ has a differential $d_A : A_n \rightarrow A_{n-1}$. A chain map is a *quasi-isomorphism* if it induces an isomorphism on homology in every degree.

Complexes have both a tensor product and a complex of maps. These constructions express homotopies in the same way that products with an interval do for spaces.

Definition 2.1. For complexes $A_\bullet, K_\bullet, B_\bullet$ of abelian groups, define the tensor product by

$$(A \otimes K)_n = \bigoplus_{p+q=n} A_p \otimes_{\mathbb{Z}} K_q, \quad d(a \otimes k) = d_A a \otimes k + (-1)^p a \otimes d_K k \quad (a \in A_p).$$

Define the *mapping complex* by

$$\underline{\text{Mor}}(A, B)_r = \prod_{n \in \mathbb{Z}} \text{Mor}_{\mathbb{Z}}(A_n, B_{n+r}), \quad \partial \phi = d_B \phi - (-1)^r \phi d_A.$$

Thus ∂ lowers degree by one. Its degree-zero cycles are chain maps, and a degree-one element s with $\partial s = f_1 - f_0$ is a chain homotopy from f_0 to f_1 .

Composition of graded maps gives a chain map

$$\underline{\text{Mor}}(B, C) \otimes \underline{\text{Mor}}(A, B) \rightarrow \underline{\text{Mor}}(A, C).$$

Indeed, for ψ of degree s , $\partial(\psi \phi) = (\partial \psi) \phi + (-1)^s \psi (\partial \phi)$. Composition is associative, and the identity maps are degree-zero cycles. Hence the morphisms between complexes themselves form complexes, compatibly with composition.

Proposition 2.2. For complexes of abelian groups, there is a natural isomorphism of chain complexes

$$\underline{\text{Mor}}(A \otimes K, B) \cong \underline{\text{Mor}}(A, \underline{\text{Mor}}(K, B)).$$

It sends a homogeneous map F to the map $F^\#$ defined by $F^\#(a)(k) = F(a \otimes k)$.

Proof. Degreewise, the isomorphism is the tensor–Mor adjunction for abelian groups, with the direct sums in the tensor product becoming products of morphism groups. For F of degree r and $a \in A_p$, the two differentials agree since

$$\begin{aligned} (\partial F)^\#(a)(k) &= d_B F(a \otimes k) - (-1)^r F(d_A a \otimes k) \\ &\quad - (-1)^{r+p} F(a \otimes d_K k) \\ &= (\partial F^\#)(a)(k). \end{aligned}$$

□

For fixed K , this says concretely that forming $A \otimes K$ is left adjoint to forming $\underline{\text{Mor}}(K, B)$, including at the level of mapping complexes.

The chain complex $\mathbb{Z}[\Delta_{\text{nd}}^1]$ plays the role of the interval. It is the normalized chain complex freely generated by the nondegenerate simplices of Δ^1 : the vertices e_0, e_1 in degree zero and the edge e_{01} in degree one, with $de_{01} = e_1 - e_0$ and no other nonzero groups. For chain maps $f_0, f_1 : A_\bullet \rightarrow B_\bullet$, a chain homotopy from f_0 to f_1 is equivalently a chain map

$$h : A_\bullet \otimes \mathbb{Z}[\Delta_{\text{nd}}^1] \rightarrow B_\bullet$$

whose restrictions to the endpoints are f_0, f_1 . Indeed, for $a \in A_n$, a degree-one map $s : A_\bullet \rightarrow B_{\bullet+1}$ determines the graded map h by

$$h(a \otimes e_0) = f_0(a), \quad h(a \otimes e_1) = f_1(a), \quad h(a \otimes e_{01}) = (-1)^n s(a).$$

With the tensor differential of Definition 2.1, the chain-map identity on $a \otimes e_{01}$ is

$$(-1)^n d_B s(a) = (-1)^{n-1} s(d_A a) + (-1)^n (f_1(a) - f_0(a)),$$

or $d_B s + s d_A = f_1 - f_0$. The identities on the two vertices say that f_0, f_1 are chain maps.

Remark 2.3. For an additive category $\mathcal{A}$, the mapping complex is defined in the same way using $\text{Mor}_{\mathcal{A}}$, with the products taken in abelian groups. Its zeroth homology gives the morphisms in the homotopy category:

$$\text{Mor}_{K(\mathcal{A})}(A_{\bullet}, B_{\bullet}) = H_0 \underline{\text{Mor}}_{\mathcal{A}}(A_{\bullet}, B_{\bullet}).$$

Indeed, cycles are chain maps and boundaries are differences of chain-homotopic maps. If $\mathcal{A}$ is abelian, the derived category is

$$D(\mathcal{A}) = K(\mathcal{A})[\text{quasi-isomorphisms}^{-1}].$$

Chain-homotopy equivalences are already invertible in $K(\mathcal{A})$; quasi-isomorphisms need not be. Thus derived morphisms are not in general computed by the unchanged mapping complex. For modules, if $B_n = 0$ for $n \gg 0$, choose a quasi-isomorphism $B_{\bullet} \rightarrow I_{\bullet}$ with $I_{\bullet}$ bounded above and termwise injective. Then

$$\text{Mor}_{D(\mathcal{A})}(A_{\bullet}, B_{\bullet}) \cong H_0 \underline{\text{Mor}}_{\mathcal{A}}(A_{\bullet}, I_{\bullet}).$$

Here $\mathcal{A}$ is the module category. This is the bounded-above injective-resolution computation in homological grading; see the Stacks Project, Lemma 13.18.2 and Lemma 13.18.8, written there in cohomological grading.

Definition 2.4. A homotopy-commutative square of complexes consists of chain maps

$$\begin{array}{ccc} A_{\bullet} & \xrightarrow{f} & B_{\bullet} \\ g \downarrow & & \downarrow i \\ C_{\bullet} & \xrightarrow{j} & D_{\bullet} \end{array}$$

and a specified chain map

$$h : A_{\bullet} \otimes \mathbb{Z}[\Delta_{\text{nd}}^1] \rightarrow D_{\bullet}, \quad h(a \otimes e_0) = if(a), \quad h(a \otimes e_1) = jg(a).$$

For $a \in A_n$, define its degree-one component by $s(a) = (-1)^n h(a \otimes e_{01})$. The chain-map identity for h is equivalent to

$$d_D s + s d_A = jg - if. \tag{1}$$

Thus h is a homotopy from if to jg , represented by the degree-one map s .

Definition 2.5. For chain maps $f : A_{\bullet} \rightarrow B_{\bullet}$ and $g : A_{\bullet} \rightarrow C_{\bullet}$, define their *double mapping cylinder* $Z_{\bullet}$ by the groups and differential

$$\begin{aligned} Z_n &= B_n \oplus C_n \oplus A_{n-1}, \\ d_Z(b, c, a) &= (d_B b + f(a), d_C c + g(a), -d_A a). \end{aligned} \tag{2}$$

Its structure maps are $i_Z(b) = (-b, 0, 0)$ and $j_Z(c) = (0, c, 0)$. The degree-one map $s_Z(a) = (0, 0, a)$ determines the homotopy

$$h_Z : A_{\bullet} \otimes \mathbb{Z}[\Delta_{\text{nd}}^1] \rightarrow Z_{\bullet}$$

with endpoints $i_Z f, j_Z g$ and $h_Z(a \otimes e_{01}) = (-1)^n s_Z(a)$ for $a \in A_n$. In mapping-cone notation, the explicitly defined complex is

$$Z_{\bullet} = \text{Cone}((f, g) : A_{\bullet} \rightarrow B_{\bullet} \oplus C_{\bullet}).$$

The letters a, b, c here denote elements of the complexes; the morphisms in the square are f, g, i, j . The identity $d_Z^2 = 0$ follows from the chain-map identities for f, g . For example, its first component is

$$d_B(d_B b + f(a)) + f(-d_A a) = d_B f(a) - f d_A(a) = 0.$$

The second component is zero by the same identity for g , and the third is $d_A^2 a = 0$.

Proposition 2.6. For a square (f, g, i, j, h) as in Definition 2.4, there is a chain map

$$\Phi_h : Z_\bullet \rightarrow D_\bullet, \quad \Phi_h(b, c, a) = -i(b) + j(c) + s(a). \quad (3)$$

Its composites with the two structure maps of Definition 2.5 are i and j .

Proof. The chain-map condition is the homotopy identity (1). For $(b, c, a) \in Z_n$, expand the two sides:

$$\begin{aligned} \Phi_h d_Z(b, c, a) &= -id_B b + jd_C c + (jg - if)(a) - sd_A a \\ &= -id_B b + jd_C c + d_D s(a) && \text{by (1)} \\ &= d_D(-i(b) + j(c) + s(a)) && \text{since } i, j \text{ are chain maps} \\ &= d_D \Phi_h(b, c, a). \end{aligned}$$

Finally, $\Phi_h(-b, 0, 0) = i(b)$ and $\Phi_h(0, c, 0) = j(c)$, giving the required composites. $\square$

Definition 2.7. A square (f, g, i, j, h) of complexes is a *homotopy pushout square* if its comparison map Φ_h in Proposition 2.6 is a quasi-isomorphism.

This is a condition in the homotopy theory of complexes with quasi-isomorphisms as equivalences. With the structure maps i_Z, j_Z, h_Z , the canonical complex $Z_\bullet$ is a homotopy pushout: its comparison map to itself is the identity. The square with lower-right complex $D_\bullet$ is a homotopy pushout when Φ_h is a quasi-isomorphism. The sign in its first structure map makes the comparison map restrict to the given maps i and j .

2.2 The Long Exact Homology Sequence

The shifted common complex occurs as a quotient of the double mapping cylinder. This quotient produces the connecting map, including its sign.

Proposition 2.8. For a span $B_\bullet \xleftarrow{f} A_\bullet \xrightarrow{g} C_\bullet$ of complexes, the double mapping cylinder of Definition 2.5 fits into an exact sequence

$$\begin{aligned} \cdots \rightarrow H_n(A) &\xrightarrow{(f_*, g_*)} H_n(B) \oplus H_n(C) \xrightarrow{k_*} H_n(Z) \\ &\xrightarrow{\delta_Z} H_{n-1}(A) \xrightarrow{(f_*, g_*)} H_{n-1}(B) \oplus H_{n-1}(C) \rightarrow \cdots, \end{aligned}$$

where $k(b, c) = (b, c, 0)$ and, for a cycle (b, c, a) in Z_n ,

$$\delta_Z[(b, c, a)] = -[a]. \quad (4)$$

Proof. There is a degreewise split short exact sequence of complexes

$$0 \rightarrow B_\bullet \oplus C_\bullet \xrightarrow{k} Z_\bullet \xrightarrow{p} A_\bullet[1] \rightarrow 0, \quad p(b, c, a) = a.$$

Here $A[1]_n = A_{n-1}$ and the shifted differential is $-d_A$, so p is a chain map. The long exact homology sequence of this short exact sequence applies.

For a cycle $a \in A_{n-1}$, lift it to $(0, 0, a) \in Z_n$. Its boundary is $(f(a), g(a), 0)$, so the connecting map from $H_n(A[1])$ to $H_{n-1}(B \oplus C)$ is (f_*, g_*) . The map induced by p sends $[(b, c, a)]$ to $[a]$ under $H_n(A[1]) = H_{n-1}(A)$. Replacing this map by its negative does not change its kernel or image, hence preserves exactness and gives (4). This sign will agree with the Mayer-Vietoris convention (f_*, g_*) followed by $j_* - i_*$. $\square$

Theorem 2.9. For a homotopy pushout square (f, g, i, j, h) of complexes, there is a long exact sequence

$$\begin{aligned} \cdots \rightarrow H_n(A) &\xrightarrow{(f_*, g_*)} H_n(B) \oplus H_n(C) \xrightarrow{j_* - i_*} H_n(D) \\ &\xrightarrow{\delta_D} H_{n-1}(A) \xrightarrow{(f_*, g_*)} H_{n-1}(B) \oplus H_{n-1}(C) \rightarrow \cdots \end{aligned} \quad (5)$$

Its connecting map is $\delta_D = \delta_Z(\Phi_{h,*})^{-1}$.

Proof. By Definition 2.7, the homomorphism $\Phi_{h,*} : H_n(Z) \rightarrow H_n(D)$ is an isomorphism in every degree. Transport the exact sequence of Proposition 2.8 along these isomorphisms. Formula (3) gives

$$\Phi_h k(b, c) = j(c) - i(b),$$

so the map into $H_n(D)$ is $j_* - i_*$. The transported connecting map is the one stated, proving (5). $\square$

A cycle $z \in D_n$ can therefore be handled as follows. Choose a cycle $(b, c, a) \in Z_n$ whose image under Φ_h is homologous to z . The cycle conditions and the image are

$$\begin{aligned} d_A a &= 0, & d_B b &= -f(a), & d_C c &= -g(a), \\ z &\sim -i(b) + j(c) + s(a). \end{aligned}$$

Then $\delta_D[z] = -[a]$. The common chain a records the boundary that has to be matched between the two pieces. Proposition 2.8 proves that this class is independent of the representative in Z_n .

Example 2.10. Take $C_* = 0$ and $g = 0$. The double mapping cylinder is $\text{Cone}(f)$. Theorem 2.9 becomes the mapping-cone sequence for $f : A_* \rightarrow B_*$, with the map $H_n(B) \rightarrow H_n(\text{Cone}(f))$ induced by the negative of the usual summand inclusion. Exactness is unchanged by that sign. Taking both B, C to be zero instead gives $Z = A[1]$ and $H_n(Z) = H_{n-1}(A)$.

Reference. J. P. May, *A Concise Course in Algebraic Topology*, University of Chicago Press, 1999, Chapter 10, §7, p. 80.